3.4.2 P-series

We begin by recalling the Integral Test from last time.

Theorem (The Integral Test): Suppose f is a continuous, positive, decreasing function on $[1,\infty)$ and let $a_n = f(n)$. Then,

1. If $\int_1^\infty f(x)dx$ is convergent, then $\sum_{n=1}^\infty a_n$ is convergent.
2. If $\int_1^\infty f(x)dx$ is divergent, then $\sum_{n=1}^\infty a_n$ is divergent.

Example 1 (you try):

(a) Use the integral test to show that the following series is convergent when the constant, p, satisfies $p > 1$: $\sum_{n=1}^\infty \frac{1}{n^p}$.

First we note that the function $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing on $[1,\infty)$. Now, we calculate that: $\int_1^\infty \frac{1}{x^p}dx = \lim_{t\to\infty}\int_1^t \frac{1}{x^p} = \lim_{t\to\infty}\left(\frac{1}{1-p}x^{1-p}\Big|_1^t\right) = \lim_{t\to\infty}\left(\frac{1}{1-p}t^{1-p} - \frac{1}{1-p}\right)$. Since $p > 1$, we know $\lim_{t\to\infty}\left(\frac{1}{1-p}t^{1-p} - \frac{1}{1-p}\right) = \frac{1}{p-1}$. Then, since $\int_1^\infty \frac{1}{x^p}dx$ is convergent, we know by the integral test that $\sum_{n=1}^\infty \frac{1}{n^p}$ is convergent.

■

(b) Use the integral test to show that the following series is divergent when the constant, p, satisfies $0 < p < 1$: $\sum_{n=1}^\infty \frac{1}{n^p}$.

First we note that the function $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing on $[1,\infty)$. Now, we calculate that: $\int_1^\infty \frac{1}{x^p}dx = \lim_{t\to\infty}\int_1^t \frac{1}{x^p} = \lim_{t\to\infty}\left(\frac{1}{1-p}x^{1-p}\Big|_1^t\right) = \lim_{t\to\infty}\left(\frac{1}{1-p}t^{1-p} - \frac{1}{1-p}\right) = \infty$. Then, since $\int_1^\infty \frac{1}{x^p}dx$ is divergent, we know by the integral test that $\sum_{n=1}^\infty \frac{1}{n^p}$ is divergent.

■

Note: We can generalize this example to form the p-series test.

Theorem (*p*-series test): The *p*-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Example 2: Determine whether the following series converges or diverges.

$$1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[3]{4}} + \ldots$$

We note that this series can be expressed as $\sum_{n=1}^{\infty} \frac{1}{n^{1/3}}$ which is a *p*-series with $p = \frac{1}{3} < 1$. Therefore, the given series diverges.

Example 3 (you try):

Determine whether the following series converges or diverges: $\sum_{k=1}^{\infty} \frac{9k^{1/2}}{2k^2}$

We note that: $\sum_{k=1}^{\infty} \frac{9k^{1/2}}{2k^2} = \frac{9}{2} \cdot \sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$. Since $\sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$ is a *p*-series with $p = \frac{3}{2}$ it converges and therefore the given series converges.

■